Unit-1 Introduction to PC Hardware

❖ Computer, Hardware ,Software

Computer: A computer is an electronic device that:

- accepts input
- processes data
- stores data
- produces output

Hardware:

- Computer Hardware is the physical part of the computer system, the machinery and equipment.
- Parts of the computer "you can see"

Firmware :

- Firmware is low-level software often stored on electrically programmable mernory devices.
- Firmware is given its name because it is treated like hardware and run by other software programs.
- Firmware is a combination of software and hardware. Computer chips that have data or programs recorded on them are firmware.
- These chips commonly include the following:
  - ROMs(read-only memory)
  - PROMs(programmable read-only memory)
  - EPROMs(erasable programmable read-only memory)

❖ Examples of Computer Hardware

Software:

Computer Software are programs that tell the computer what to do.

Examples :

- Microsoft Word-word processing program
- Microsoft PowerPoint-presentation program

- Microsoft Excel-work book program used to track, calculate, and analyze numeric data

❖ BASIC PART OF COMPUTER SYSTEM

- **Input Devices:** devices that input information into the computerϖ such as a keyboard, mouse, scanner, and digital camera.

- **Output Devices:** devices that output information from the computer such as a printer and monitor.

- **CPU (Central Processing Unit)** also called the Microprocessor or **"The Brain"** of the Computer.
- **Processor speed:** The speed at which a microprocessor executes instructions. This is usually measured in megahertz (MHz).
- **Brands of Processors include:**
  - Pentium
  - Celeron
  - MAC
  - AMD
  - Cyrix

❖ Types of Computer System OR Classification of Computer.

1) Personal Computer (Micro Computer):-

- a small computer designed for use by a single user at a time.
- A PC or microcomputer uses a single chip (microprocessor) for its central processing unit (CPU).
- "Microcomputer" is now primarily used to mean a PC, but it can refer to any kind of small computer, such as a desktop computer, laptop computer, tablet, smartphone, or wearable.
- PC has low cost and smaller size.
- It is useful for individuals without intermediate compter operator.
- The most common operating system such as Microsoft Window, Mac OS , UNIX, LINUX etc are used in PC.
- It can be capable to connect with other PC on LAN.

➢ **<u>Types of personal computers</u>**

I. **Desktop computer** – a personal computer that is designed to stay at one location and fits on or under a desk. It typically has a monitor, keyboard, mouse, and a tower (system unit).

II. **Laptop computer (or notebook) –** A portable personal computer that is small enough to rest on the user's lap and can be powered by a battery. It includes a flip down screen and a keyboard with a touchpad.

III. **Tablet –** A wireless touch screen PC that is slightly smaller and weighs less than the average laptop.

VPMP POLYTECHNIC

IV. **Smartphone** – A mobile phone that performs many of the functions of a personal computer.

2) Mainframe Computer:-

- A high-performance computer used for large information processing jobs.
- The Processing power of the mainframe is time- shared between all of the users.
- They are used in situations where a company wants the processing and information storage in a centralized location

- Mainframe computers are primarily used in :
    - Institutions
    - Research
    - Academics
    - Health care
    - Libraries
    - Large businesses

- Financial institutions
- Stock brokerage firms
- Insurance agencies

- Mainframe computers are useful for tasks related to:
  - Census taking
  - Industry and consumer statistics
  - Enterprise resource planning
  - Transaction processing

3) Minicomputers:-

- a mid-range computer that is intermediate in size, power, speed, storage capacity, etc., between a mainframe and a microcomputer.

- Minicomputers are used by small organizations.

- "Minicomputer" is a term that is no longer used much. In recent years, minicomputers are often referred to as small or midsize servers (a server is a central computer that provides information to other computers).

4) Workstation Computer:-

- A Workstation is a Powerful , high-end microcomputer.
- They contain one or more microprocessor CPUs.
- They may be used by a single-user for applications requiring more power than a typical PC.
- The term " workstation" also has an alternate meaning: In networking, any client computer connected to the network that accesses server resources may be called a workstation.

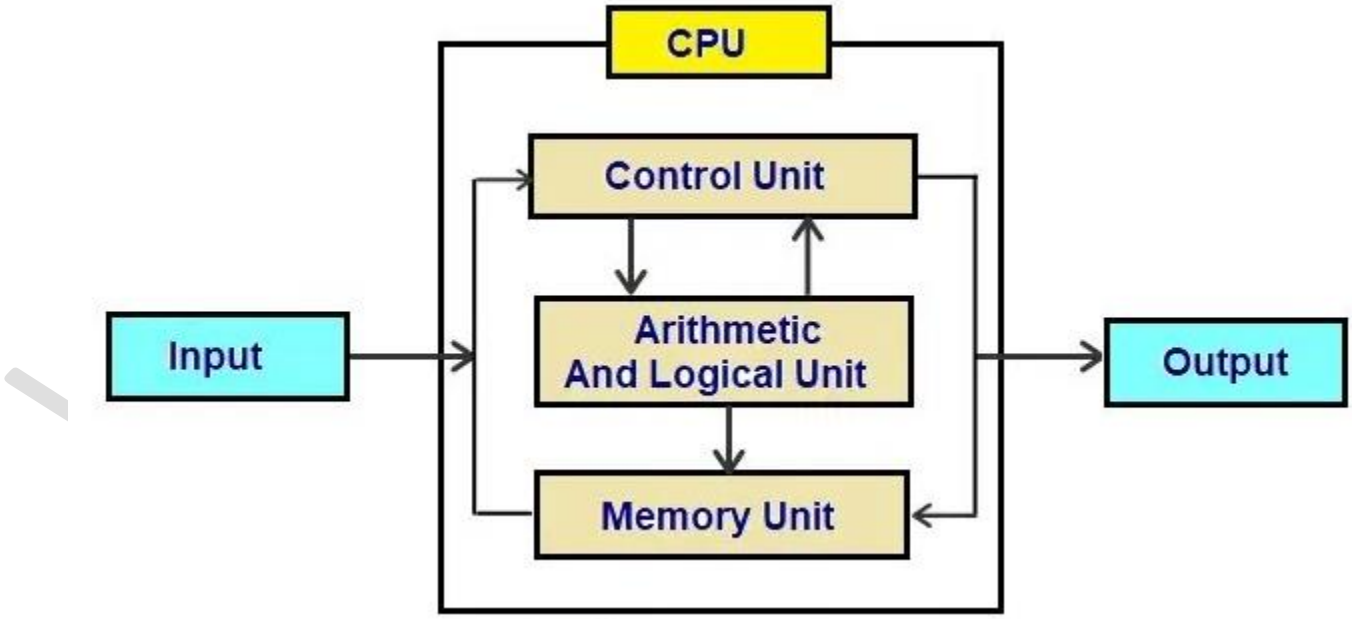
5) Super Computer:-

- a powerful computer that can process large amounts of data and do a great amount of computation very quickly.
- Supercomputers are used for areas related to:
  - Science
  - Engineering
  - Education
  - Defense
  - Aerospace

- Supercomputers are useful for applications involving very large databases or that require a great amount of computation.

- Supercomputers are used for complex tasks, such as:
  - Weather forecasting
  - Climate research
  - Scientific simulation
  - Oil and gas exploration
  - Quantum mechanics
  - Cryptanalysis

❖ Block Diagram of PC



- **Input:-**

- All the data received by the computer goes through the input unit. The input unit comprises different devices like a mouse, keyboard, scanner, etc. In other words, each of these devices acts as a mediator between the users and the computer.
- The data that is to be processed is put through the input unit. The computer accepts the raw data in binary form. It then processes the data and produces the desired output.
- The 3 major functions of the input unit are-
- Take the data to be processed by the user.
- Convert the given data into machine-readable form.And then, transmit the converted data into the main memory of the computer. The sole purpose is to connect the user and the computer. In addition, this creates easy communication between them.

- **CPU – Central Processing Unit**

- Central Processing Unit or the CPU, is the brain of the computer. It works the same way a human brain works. As the brain controls all human activities, similarly the CPU controls all the tasks.
- Moreover, the CPU conducts all the arithmetical and logical operations in the computer.
- Now the CPU comprises of two units, namely – ALU (Arithmetic Logic Unit) and CU (Control Unit). Both of these units work in sync.

- **ALU – Arithmetic Logic Unit**

- The Arithmetic Logic Unit is made of two terms, arithmetic and logic. There are two primary functions that this unit performs.
- Data is inserted through the input unit into the primary memory. Performs the basic arithmetical operation on it. Like addition, subtraction, multiplication, and division. It performs all sorts of calculations required on the data. Then sends back data to the storage.
- The unit is also responsible for performing logical operations like AND, OR, Equal to, Less than, etc. In addition to this it conducts merging, sorting, and selection of the given data.

- **CU – Control Unit**

- The control unit as the name suggests is the controller of all the activities/tasks and operations. All this is performed inside the computer.
- The memory unit sends a set of instructions to the control unit. Then the control unit in turn converts those instructions. After that these instructions are converted to control signals.
- These control signals help in prioritizing and scheduling activities. Thus, the control unit coordinates the tasks inside the computer in sync with the input and output units.

- **Memory Unit**

- All the data that has to be processed or has been processed is stored in the memory unit. The memory unit acts as a hub of all the data. It transmits it to the required part of the computer whenever necessary.
- The memory unit works in sync with the CPU. This helps in faster accessing and processing of the data. Thus, making tasks easier and quicker.
- **There are two types of computer memory-**

1) **Primary memory** – This type of memory cannot store a vast amount of data. Therefore, it is only used to store recent data. The data stored in this is temporary. It can get erased once the power is switched off. Therefore, is also called temporary memory or main memory.

   - RAM stands for Random Access Memory. It is an example of primary memory. This memory is directly accessible by the CPU. It is used for reading and writing purposes. For data to be processed, it has to be first transferred to the RAM and then to the CPU.

2) **Secondary memory** – As explained above, the primary memory stores temporary data. Thus it cannot be accessed in the future. For permanent storage purposes, secondary memory is used. It is also called permanent memory or auxiliary memory. The hard disk is an example of secondary memory. Even in a power failure data does not get erased easily.

➢ **Output**

- There is nothing to be amazed by what the output unit is used for. All the information sent to the computer once processed is received by the user through the output unit. Devices like printers, monitors, projectors, etc. all come under the output unit.
- The output unit displays the data either in the form of a soft copy or a hard copy. The printer is for the hard copy. The monitor is for the display. The output unit accepts the data in binary form from the computer. It then converts it into a readable form for the user.

❖ List and identify components of PC / laptop and specify its importance.

➢ **There are several basic components of computer hardware including:**

- Central processing unit (CPU)
- Motherboard
- Memory Module
- Video graphics array port
- Power supply
- Cooling fan
- Hard drive
- Monitor

1) **CPU :-**

- Central Processing Unit or the CPU, is the brain of the computer.
- The microprocessor is a single integrated circuit (IC) , which containing sophisticated circuitry that allows it to perform simple arithmetic calculation to more complex program controlling.
- The two main part of the microprocessor are the bus interface unit and the execution unit.
- The bus interface unit does data transfer to all the circuit on the system.
- The execution unit is the part that actually carries out the instructions.

2) Motherboard :-

- The motherboard is the main circuit of the PC.
- It has various electronic components and chips which are linked together by metal tracks called traces.
- In addition to this components and chips motherboard may contain extra sockets for optional items.
- A socket is soldered into the circuit board and contains holes into which you can plug a chip's pins.

3) Memory Module :-

- There are the different types of memory in system.
- The most common are ROM, RAM, and Cache memory.
- The ROM is used for storing the small program like BIOS.
- The RAM used for storing temporary program and data for the CPU.
- The Cache memory is very costly and fast memory generally used for storing very small amount of information.

4) Video graphics array (VGA) :-

- Short for video graphics adapter or video graphics array.
- VGA is a popular display standard developed by IBM and introduced in 1987.
- VGA provides 640 x 480 resolution color display screens with a refresh rate of 60 Hz and 16 colors displayed at a time. If the resolution is lowered to 320 x 200, 256 colors are shown.
- VGA utilizes analog signals, which means it's only capable of lower resolutions and a lower quality display on screens.

5) Power supply :-

- The power supply generally located in the right corner of a desktop PC.
- It converts the AC power to DC and creates the various voltages needed to power your computer's devices.
- The power supply has several important rating like input voltage, frequency, output voltage etc.

6) Cooling fan :-

- A cooling fan is any fan inside, or attached to, a computer case used for active cooling.
- Fans are used to draw cooler air into the case from the outside, expel warm air from inside and move air across a heat sink to cool a particular component.

7) Hard drive :-

- A hard drive is the hardware component that stores all of your digital content.
- Your documents, pictures, music, videos, programs, application preferences, and operating system represent digital content stored on a hard drive.

8) Monitor :-

- The monitor are used as an output device for display text and graphic.

- The monitor looks like a television screen.
- It is connected to the main computer box.
- There are several types of monitors, but they all function in the same way.
- The power button is usually located on the front of the monitor, beneath the screen.
- There will be a small light to indicate if the power is on.
- Monitor is available in different size and resolution.
- They can build using different technology like CRT, LCD etc.

❖ Prepare latest specification of PC / laptop

Component	Specification
Processor:	11th, 12th or 13th Gen Intel Core i5, i7 or i9 Processor, Apple Silicon M1, M1 Pro, M1 Max, M1 Ultra or M2 Processor (CPU)
Operating System:	Microsoft Windows 10 or Windows 11 Home, Pro, Enterprise or Education version *or* macOS 11.X "Big Sur" or macOS 12.X "Monterey."
Office Suite:	Microsoft Office 365 for Windows or Macintosh. Microsoft Office 365 is available for no-cost download for all enrolled students. See the UDel Office 365 for Education services page for details.
Memory (RAM):	8-16 GB of RAM
Storage:	240 GB solid state drive, or larger.
Video/Graphics:	Integrated or discrete graphics processor capable of 1440 X 900 resolution, or better (1920 X 1080 *or* 1200 ideal).
Monitor:	for notebook: 13″ – 17″ display for desktop: 24″ – 27″ widescreen flat-panel display
Mouse:	Built-in or external track pad, wireless and/or USB, 2-button, optical mouse
Sound:	Built-in audio or soundcard, and speakers
Headphones:	Headphones or ear buds, with built-in microphone
Webcam:	Either external USB device or built-in
Network:	802.11ac (or better) Wi-Fi capability.
Warranty:	3 year on-site (better) or depot warranty, accidental damage service highly recommended.
Backup device/scheme:	External USB or Thunderbolt drive with Time Machine (Mac) or Windows backup software and/or cloud backup service/subscription.

❖ Explain and List out of identify various types of port found in PC / laptop ?

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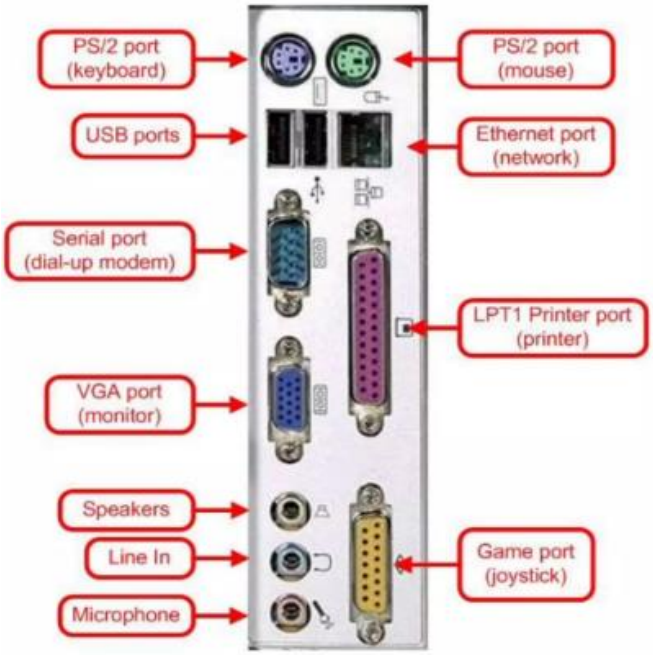
- A computer is a device that transforms data into meaningful information. It processes the input according to the set of instructions provided to it by the user and gives the desired output. As we know that we can connect multiple external devices with the computer system. Now, these devices are connected with the computer using Ports. The ports are the physical docking points present in the

computer through which the external devices are connected using cables. Or in other words, a port is an interface between the motherboard and an external device of the computer.

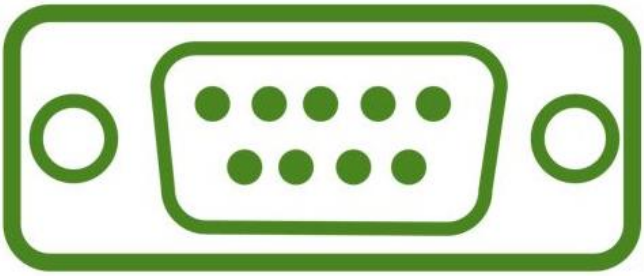


- **There are different types of ports available**:

1) Serial port
2) Parallel port
3) USB port
4) PS/2 port
5) VGA port
6) Modem port
7) FireWire Port
8) Sockets
9) Infrared Port
10) Game Port
11) Digital Video Interface(DVI) Port
12) Ethernet Port
13) Sound port

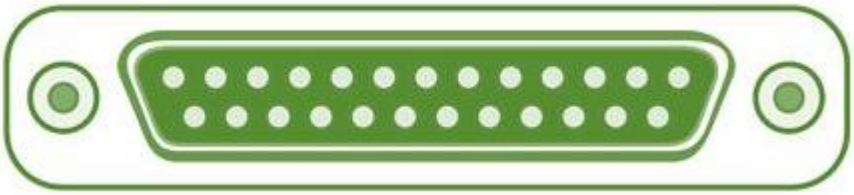
1) Serial Port:-

- A serial port is also called a communication port and they are used for connection of external devices like a modem, mouse, or keyboard (basically in older PCs).
- Serial cables are cheaper to make in comparison to parallel cables and they are easier to shield from interference.
- There are two versions of it, which are 9 pin model and 25 pin model. It transmits data at 115 KB/sec.



2) Parallel Port:-

- Parallel ports are generally used for connecting scanners and printers.
- It can send several bits at the same time as it uses parallel communication.
- Its data transfer speed is much higher in comparison with the serial port.
- It is a 25 pin model.
- It is also known as Printer Port or Line Printer Port.



3) USB (Universal Serial Bus)Port:-

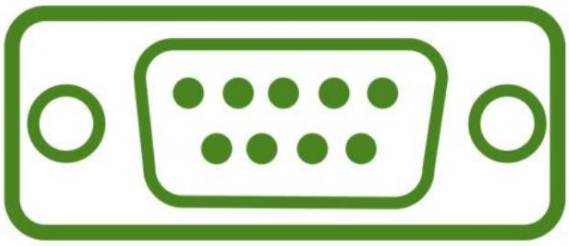
- In 1997 USB was first introduced. This can connect all kinds of external USB devices, like external hard disk, printer, scanner, mouse, keyboard, etc.
- There are minimum of two USB Ports provided in most of the computer systems.

4) PS/2 port :-

- PS/2 ports are special ports used for connecting old computer keyboard and mouse.
- It was invented by IBM. In old computers, there are minimum of two PS/2 Ports, each for the keyboard and the mouse.
- It is a 6 pin mini Din connector.

5) VGA port:-

- VGA ports also known as Video Graphic Array connector are those which connect the monitor to a computer's video card.
- VGA port has 15 holes and it is similar to the serial port connector. But VGA Ports have holes in it and the serial port connector has pins in it.

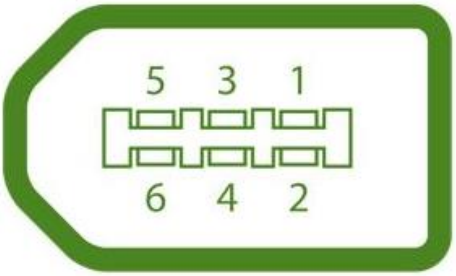


6) Modem Port:-

- As the name suggests, a Modem port is used to connects a PC's modem to the telephone network.

7) FireWire (IEEE 1394 interface) Port:-

- The IEEE 1394 interface, which is developed in the late 1980s and early 1990s by Apple as FireWire.
- It can transfer large amount of data at very high speed.
- It is used to connect camcorders and video equipment to the computer.
- It comes up with three variants which are 4-Pin FireWire 400 connector, 6-Pin FireWire 400 connector, and 9-Pin FireWire 800 connector.



8) Sockets:-

- Microphones and speakers are connected with the help of Sockets to the sound card of the computer.

9) Infrared Port:-

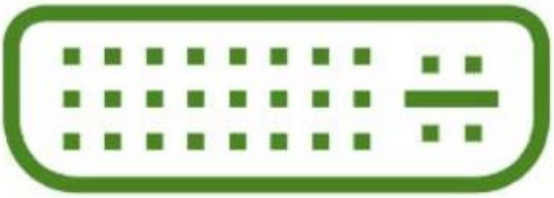
- An Infrared(IR) port is used to sends and receives infrared signals from other devices. It is a kind of wireless type port with a limited range of 5-10ft.

10) Game Port:-

- These ports are used previously to connect a joystick to a PC. But nowadays it is replaced by USB ports.

11) Digital Video Interface(DVI) Port :-

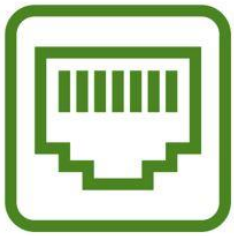
- DVI Port is used to connects LCD(flat panel) monitor to the computer's high-end video graphic cards and it is very popular among video card manufacturers.



12) Ethernet Port:-

- Ethernet Port helps to connect to a network and high-speed Internet(provided by LAN or other sources).

- It connects the network cable to a computer and resides in a Ethernet card.
- It provides a data travel speed of 10 Mb to 1000 Mb(megabits) per second.



13) Sound Port:-

- Connect microphone, speakers to sound card of the computer.

❖ Explain What are the connectors? And list out the types of connectors.

- A connector is a physical interface that enables communication between devices or between a device and a cable.
- Connectors allow the transmission of power and data signals through contact points called pins.
- The connector fits into a port, which is the socket or entrance where the connector is inserted.
- Connectors provide both a mechanical and electrical interface between devices. On a mechanical level, they ensure proper alignment and a secure connection. Electrically, they enable the flow of power and data signals across devices. Common types of connectors include USB, HDMI, Ethernet, and audio jacks.
- Each port type serves a distinct purpose and uses specific connectors to link with other devices.
- Network-enabled devices will typically have multiple ports to support varied connectivity needs.
- The configuration of ports also differs based on device type. For example, laptops have different port arrangements compared to desktop computers.

- **USB Ports and Connectors**
- USB (Universal Serial Bus) has become the standard for connecting peripherals and accessories to computers and mobile devices. Cable USB uses simple, compact connectors that offer plug-and-play convenience.
- Common types of USB connectors include:

- **USB-A**
- The standard rectangular USB connector that plugs into USB host ports on computers. USB-A is compatible with USB 1.0, 2.0 and 3.0.

- **USB-B**
- Rectangular connectors for the device end of USB cables. Used for printers, scanners and other peripherals.

- **USB-C**
- Oval-shaped, reversible USB connector for both hosts and devices. Supports USB 3.1 and Thunderbolt 3.
- Mini and Micro USB
- Smaller versions of USB-B commonly used in mobile devices, cameras and portable hubs.

➢ **Common USB connector types**

- The versatility of USB makes it a go-to choice for diverse peripherals. USB connectors are designed to withstand frequent insertion and removal without damage. However, always match the connector shape and USB version compatibility to avoid issues.

➢ **Video and Audio Connectors**

- Connections for audio and video have specialized connectors tailored to their signal requirements. For video, common connectors include:

➢ **video and audio connectors**

- **HDMI**
- **HDMI (High Definition Multimedia Interface) connectors** are slim, compact connectors used for HDTVs and other HD-capable video devices. HDMI provides high-quality digital video and audio transmission through a single cable.
- **Display Port**

- Display Port connectors can transmit very high resolutions including 4K and 8K video. They provide higher bandwidth performance than HDMI connectors.
- **VGA**
- VGA (Video Graphics Array) uses analog signals. The 15-pin DE-15 connector has screws to securely fasten to a VGA port.

➢ **For audio, two popular options are:**

- **3.5mm audio jack**
- Also known as a headphone jack or aux port. The 3.5mm TRS connector is compatible with most headphones, speakers, microphones and mobile devices.
- **Optical audio**
- Optical connectors use light to transmit digital audio signals. The TOSLINK optical connector has a square tip made of plastic fiber.

➢ Types of Connectors

- There are many different types of connectors available that serve various functions. Let's take a look at some of the most common varieties:-

1) Ethernet Cable Connectors

- Ethernet cables are used to connect devices to networks and the internet. There are two main types of Ethernet connectors:

- **RJ45 –** The RJ45 connector looks like a large phone jack. It has 8 pins inside that correspond with the 8 wires in an Ethernet cable. RJ45 is the standard connector for Ethernet networks.
- **RJ11** – RJ11 connectors have 6 pins and are only capable of handling two pairs of wires. They are too small for full Ethernet connections but may be used for telephone and DSL internet connections.

2) Coaxial Cable Connectors

- Coaxial cables are often used for cable TV and internet connections. Common coax connectors include:
- **F-type** – The F-type connector has a screw-on design that provides a solid connection for TV and cable internet. They are simple to install even for novices.
- **BNC** – BNC (Bayonet Neill-Councilmen) connectors are used for higher frequency connections and applications where a quick disconnect is required. They twist and lock into place.
-

- **RCA** – RCA connectors consist of a central pin surrounded by a cylindrical conductor. They are commonly yellow for video and white/red for stereo audio.

3) Fiber Optic Cable Connectors

- Fiber optic cables transmit data as light pulses along glass or plastic fibers. Some types of fiber optic connectors include:
- **LC** – The LC connector is a small form factor connecter half the size of an RJ45. It has a locking tab for a reliable connection.
- **SC** – SC connectors are snap-in connectors that click into place for fast locking. They are commonly used for high speed telecom networks.
- **ST** – ST connectors were an early fiber optic connector design with a bayonet twist-locking mechanism. They have been mostly phased out in favor of smaller designs.

❖ Compare between Port and Connectors.

Port	Connectors
1) Sockets on devices where cables plugin	**1)** Plugs on the end of cables that connect to ports
2) Allow network connections	**2)** Physically join devices over networks
3) Fixed on devices	**3)** Can be removed and changed
4) Standard shapes and sizes	**4)** Vary based on port compatibility
5) Examples: Ethernet, USB, HDMI	**5)** Examples: RJ45, USB-A, HDMI